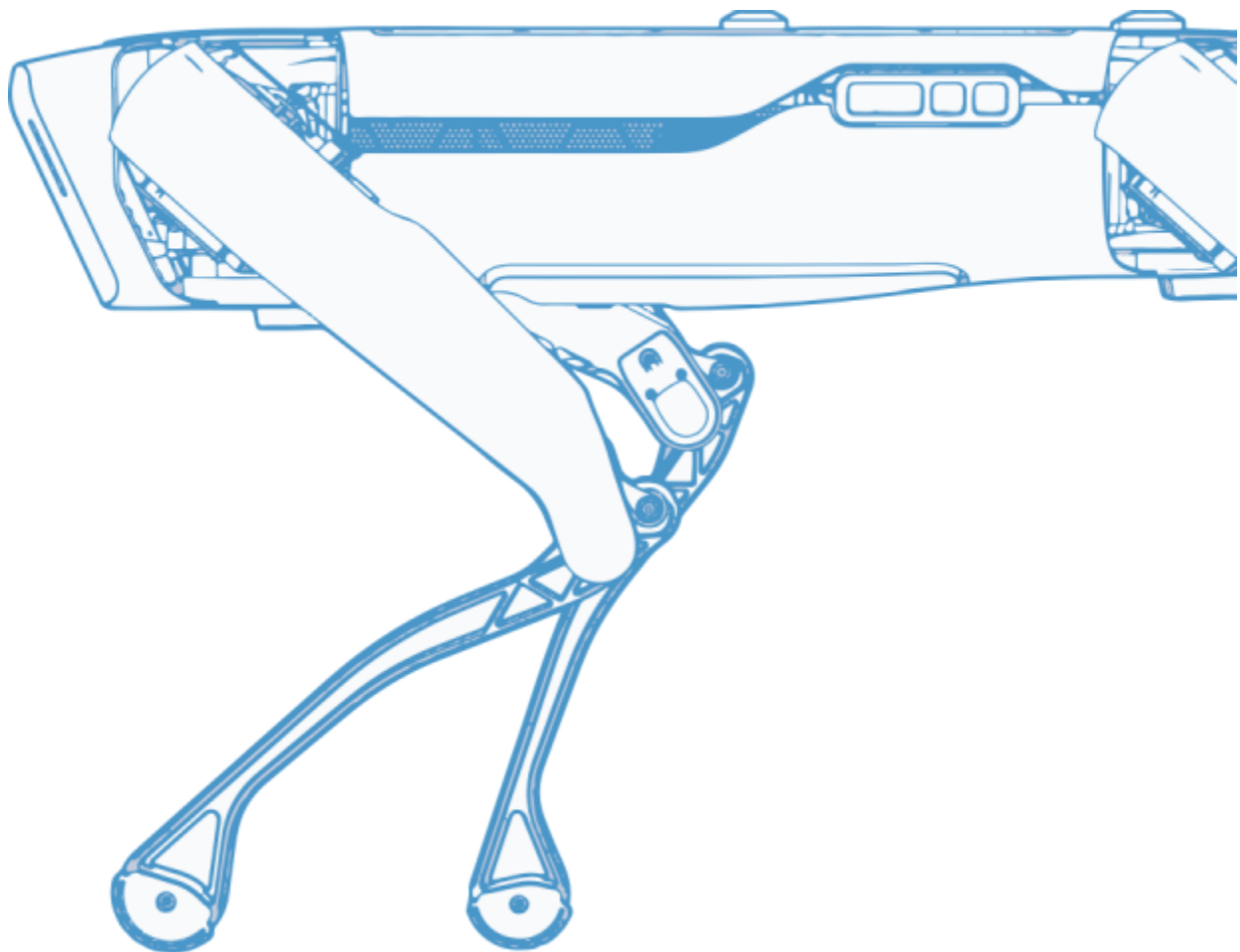


Spot + Spot Arm Information for Use (IFU)

v1.0 Original instructions



Boston Dynamics, Inc.

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Version 1.0

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8.1 EU Declaration of Incorporation

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1 Introduction

This document contains critical safety information for Spot Arm.

Responsible use of the Spot Arm is crucial to prevent dangerous conditions for operators and others nearby. Make sure to read, understand and comply with this document to lessen the risk of injuries or damage to the robot or other property.

Please read the entire document before using the Spot Arm and keep the document in a readily accessible location afterwards.

Additional information, including complete documentation on the Spot robot and Spot Dock is available on the [Boston Dynamics Support Center](#).

1.1 Manufacturer information

Spot and the Spot Arm are manufactured by:

Boston Dynamics, Inc.
200 Smith Street
Waltham, MA 02451
USA

1.2 Definition of a Spot User

In the scope of the present document, a “User” is the entity that uses Spot Arm for the intended use and is responsible for the personnel associated with operation of the robot. Users are subject to workplace safety regulations and are active counterparts of the manufacturer in implementing residual risk reduction measures.

1.3 Definition of a Spot Operator

In the scope of the present document, an “Operator” is any person operating, maintaining, cleaning, or troubleshooting Spot Arm under the responsibility of a User, including person(s) who can be reasonably expected to be near Spot Arm, even if they are not directly operating the robot. Operators must be trained in the proper use of Spot Arm.



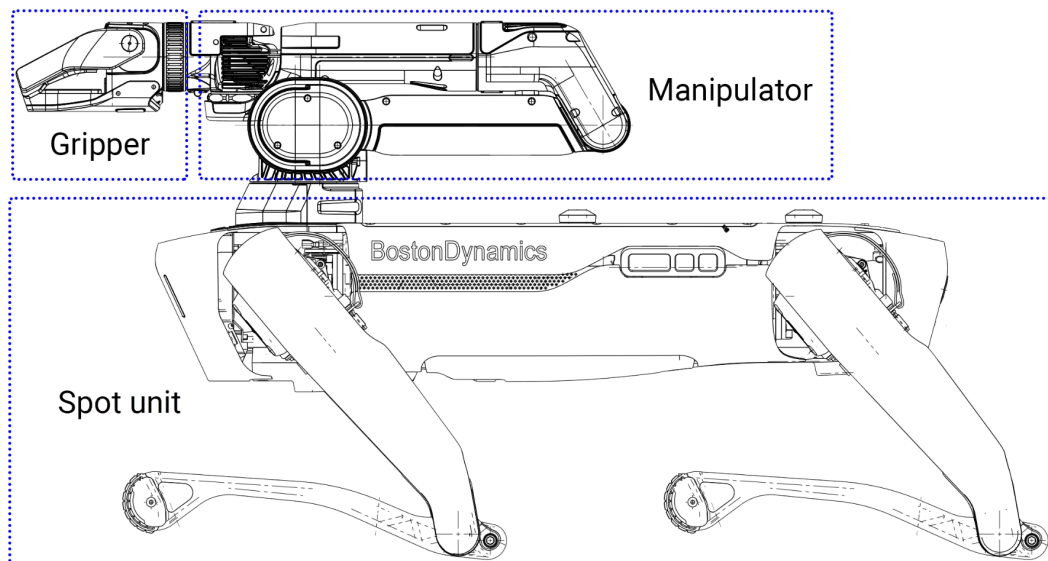
1.4 Description of the machinery

Spot + Spot Arm provides manipulation capability to the base Spot platform. The manipulator provides 6 degrees of freedom. The end-effector or gripper has a built-in single degree of freedom.

Instructions in this document primarily refer to the combined Spot + Spot Arm robot, unless otherwise specified.

1.4.1 Properties of the Spot + Spot Arm robot

- The robot performs locomotion, navigation and manipulation in a variety of indoor and outdoor environments.
- The Spot unit (quadruped part) uses embedded sensors to detect the surrounding environment.
- The Spot Arm (manipulator part) has no detection sensors. Spot + Spot Arm uses a complex set of control algorithms to manipulate objects while balancing and moving around.
- Spot + Spot Arm can be expanded with payloads.
- Spot + Spot Arm does not carry persons.
- Spot + Spot Arm is operated remotely without a physical connection, such as a cable between control units and the machinery.



Major Spot + Spot Arm components

1.5 Robot specifications

Specifications for the Spot Arm manipulator:

Spot Arm manipulator	Value
Degrees of freedom	6 + gripper
Length at full extension	985 mm
Lift capacity*	Up to 11 kg
Continuous lift capacity at 0.5 m extension*	5 kg
Operating temperature	-20°C to 45°C
Ingress protection	IP54

* At 22°C

Specifications for the Spot Arm gripper:

Spot Arm gripper	Value
Depth of grip	90 mm
Maximum aperture	175 mm
Peak clamp force (at tip of opening)	130 N
Integrated sensors	ToF, 4K RGB
Accessory port	Gigabit Ethernet, 50W power, camera sync (PPS)

The combined weight of the Spot Arm (manipulator and gripper) is 8 kg.



Specifications for Spot (quadruped part):

Category	Specification	Value
Dimensions	Length	1100 mm
	Width	500 mm
	Normal stand height (with Spot Arm stowed)	830 mm
	Normal stand height (with Spot Arm extended straight up)	1700 mm
	Tall stand height (with Spot Arm extended straight up)	1820 mm
	Net weight (with battery)	32.5 kg
	Degrees of freedom	12
Environment	Max. speed	1.6 m/s
	Ingress protection	IP54
	Operating temperature	-20°C to 45°C
Power	Slopes	+/- 30°
	Battery capacity	605 Wh
	Max. battery voltage	58.8V
	Typical runtime	60 minutes
	Standby time	110 minutes
	Charger power	400W
	Max. charge current	7A
	Time to charge	Approximately 2 hours
Payload	Battery weight	4.18 kg
	Max. payload weight with Spot + Spot Arm	6 kg
	Max. power per port	150W

	Payload ports	2
	T-slot rail bolt size	M5 x 1.0
Cameras	Camera type	Projected stereo
	Field of view	360 degrees
	Operating range	4 m
Connectivity	802.11	Explorer Wi-Fi: 2.4GHz 802.11 b/g/n Enterprise Wi-Fi: Explorer + 5GHz 802.11ac
	Ethernet	1000Base-T

1.5.1 Spot Arm workspace

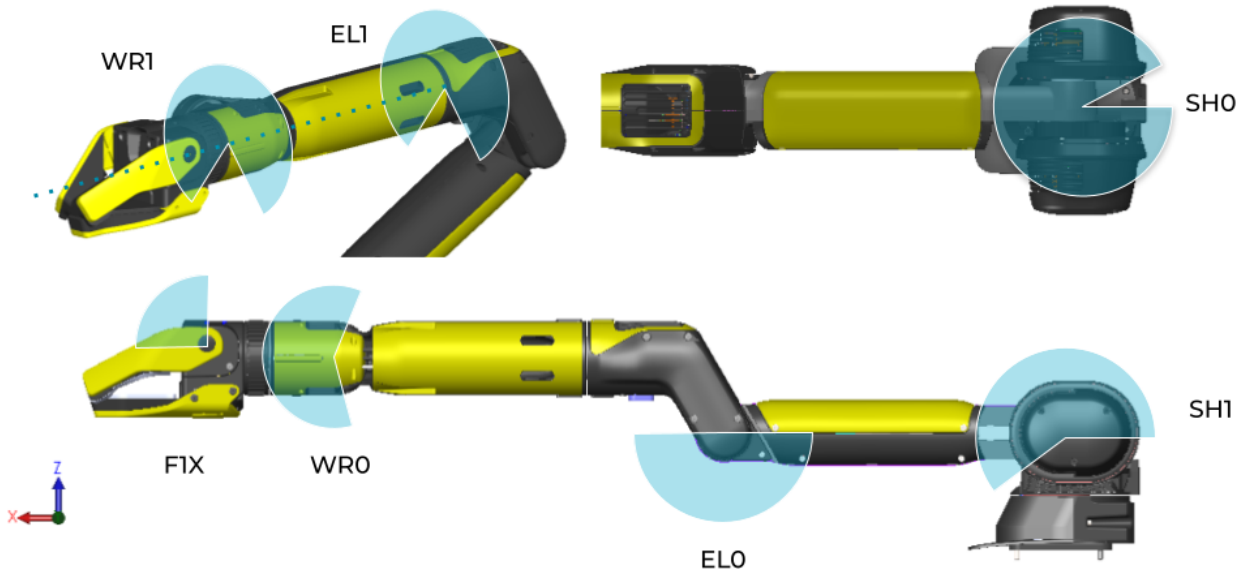
The working space of Spot Arm is determined by the volume used to plan and execute a trajectory, which depends on the combination of the following factors:

- Volume (maximum space) where the manipulator can move
- Displacement of the base

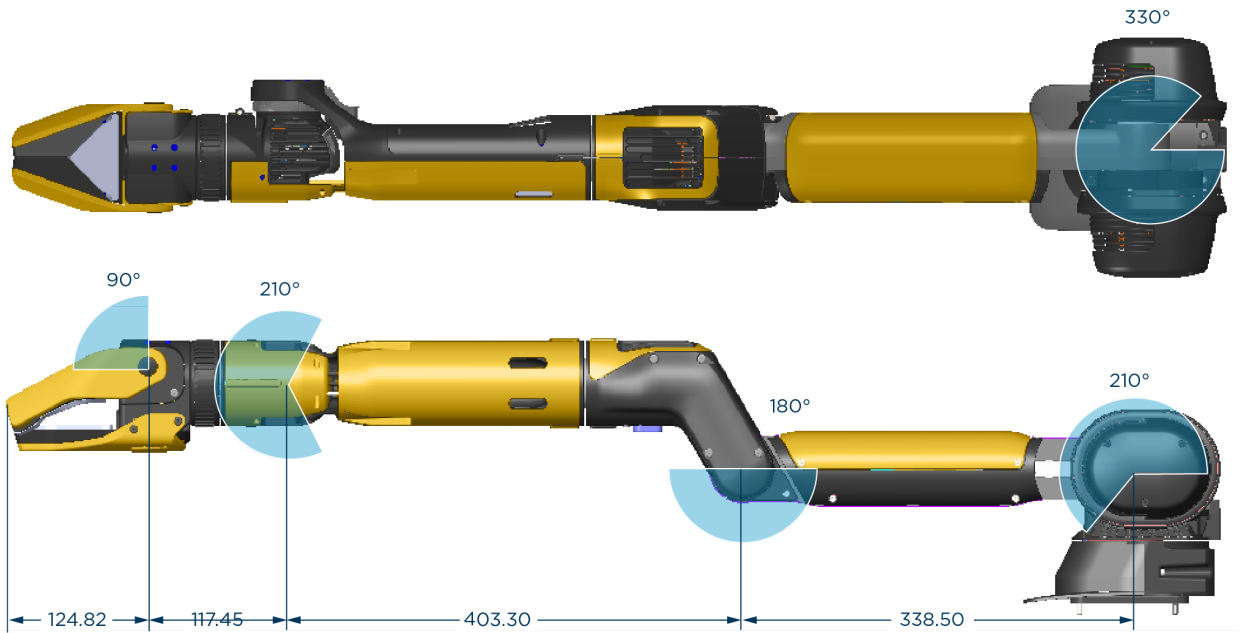
The maximum space of the manipulator alone is determined by the range of motion of the joints, as set forth below:

Joint Axis	Range	Max. velocity
SH0	-150°/180°	10 rad/s
SH1	-180°/30°	10 rad/s
EL0	0°/180°	10 rad/s
EL1	-160°/160°	10 rad/s
WR0	-105°/105°	10 rad/s
WR1	-165°/165°	10 rad/s

The length of the manipulator at full extension, including the gripper, is 985 mm. The maximum height of the Spot Arm with Spot in tall stand pose is 1820 mm.



Spot Arm joint axis designations



Spot Arm joint ROM and length of links (in millimeters)

2 Product safety overview

Spot + Spot Arm uses sophisticated control methodology for dynamically balancing while manipulating objects.

Spot + Spot Arm's behavior while in motion can be variable or unexpected with respect to planned trajectories or movements. Spot + Spot Arm behavior is affected by physical contact and the exchange of forces with manipulated objects and equipment.



This section contains important safety information, which all Users must read and understand before using Spot + Spot Arm.

It is important to read ALL parts of this documentation:

1. As an integral part of Operator training
2. Before designing and commissioning Spot + Spot Arm applications

2.1 Important safety notice

Spot + Spot Arm is a partly completed machinery¹ and the use and safety of the robot depends on specific applications to be implemented by the User including, but not limited to, tools and parts manipulated, equipment that is part of the application, and safeguards that may be integrated.

Spot + Spot Arm combines complex walking and balancing with full spatial manipulation, resulting in a wide range of motion.

Users must conduct a full risk assessment of their application prior to use. The risk assessment shall consider the entire lifecycle of the Spot + Spot Arm application, including but not limited to:

- Setting up and commissioning the Spot + Spot Arm, with particular attention to training activities.
- Regular use of the Spot + Spot Arm application, with particular attention to manipulation and navigation actions.

¹ According to the EU Machinery Directive 2006/34/CE. Although not recognized in other jurisdictions, the notion of partly completed machinery is valid in general and Spot Arm is only fully determined as complete machinery when integrated in specific applications.



- Maintenance and troubleshooting.

NOTICE

The User is responsible for implementing and maintaining risk reduction measures for specific Spot + Spot Arm applications.

2.2 Intended use

Spot + Spot Arm is intended to be used for general purpose manipulation driven by legged movements in industrial or controlled environments. These environments are subject to limitations of conditions and preparation by the Users after a survey of the expected or planned environment.

Spot + Spot Arm is not intended for collaborative applications involving human contact or proximity to humans.

Spot + Spot Arm should only be used in dedicated areas where access is restricted or where demarcation is made clear to Operators.

Spot + Spot Arm is capable of carrying additional payloads. Due to variations in the size and purpose of different payloads, added payloads may interfere with the manipulator.



Users must always identify the intended use of their final applications and conduct full risk assessments prior to use.

2.3 Misuse

The following behaviors and uses of Spot + Spot Arm by Users and Operators are prohibited:

- Using the robot before reviewing a specific application and performing a risk assessment.
- Using the robot in home environments.
- Using the robot in personal care, medical treatment, or life-critical applications.

- Entering or remaining inside a restricted area designated for the robot operations other than for the purpose of setting or recording missions.
- Transporting persons or animals.
- Transporting hazardous materials or substances.
- Intentionally harming any person with the robot or payloads mounted on the robot.
- Using the robot for any illegal purpose.
- Using the robot as a climbing aid.
- Using the robot in potentially explosive environments.
- Using the robot outside of the controlled or restricted environment.

2.4 Risk assessment

Spot + Spot Arm is a versatile machine capable of complex movements. Mechanical hazards, in particular, depend on specific applications and the robot's posture.

A risk assessment must be conducted before Spot + Spot Arm is used for the identified application for the first time.

Users must identify significant hazards in their application relating to:

- The intended manipulation tasks.
- The environment where Spot + Spot Arm is moving during manipulation or navigation.
- The parts and equipment used during operation.

Users must consider the following factors when estimating the risks of potential contacts with Spot + Spot Arm or other elements of the application:

- The severity of potential collisions that could cause injury.
- The frequency of the foreseeable occurrence of a hazard.
- The probability of errors or failures.
- The potential for collisions.
- Awareness of the surroundings.
- The speed of possible hazards.

Users must consider the intended use of Spot + Spot Arm, accidental events, and reasonably foreseeable misuse in an application.



2.4.1 Hazards

Boston Dynamics has identified the following significant hazards, which must be considered by the User when conducting a risk assessment:

- Bruising on the lower body due to contact with any part of Spot + Spot Arm after general falls and loss of balance on elevated or inclined surfaces.
- Items ejected from the gripper.
- Tripping, dragging, and entanglement of elements of the application with Spot Arm.
- Trapping of Spot Arm in equipment or parts of the layout while the Spot unit is still partly or completely free.
- Bruising upon impact with the Spot Arm after a tip-over.
- Bruising upon impact with the Spot Arm as a result of recoiling movements of Spot + Spot Arm.
- Cutting, puncturing, or bruising by items manipulated with Spot Arm.
- Bone injury due to contact with heavy parts of the equipment involved in Spot Arm manipulation.

The foregoing list is not exhaustive, as other significant hazards can be present depending on the particular Spot + Spot Arm application.

In particular, Users should review the environmental conditions and the properties of their layouts and determine whether their application causes or presents additional hazards.

2.4.2 Estimating the risk of pinch points

Pinch points are present at:

- The gripper tip
- Along the manipulator (at the wrist, elbow, and shoulder joints)
- Between the shoulder motors and the top of the front hips, specifically when the robot base is rolling laterally
- Hips and knees



Spot + Spot Arm pinch point hazards

No intentional accesses to powered moving joints should be part of any intended use.

NOTICE

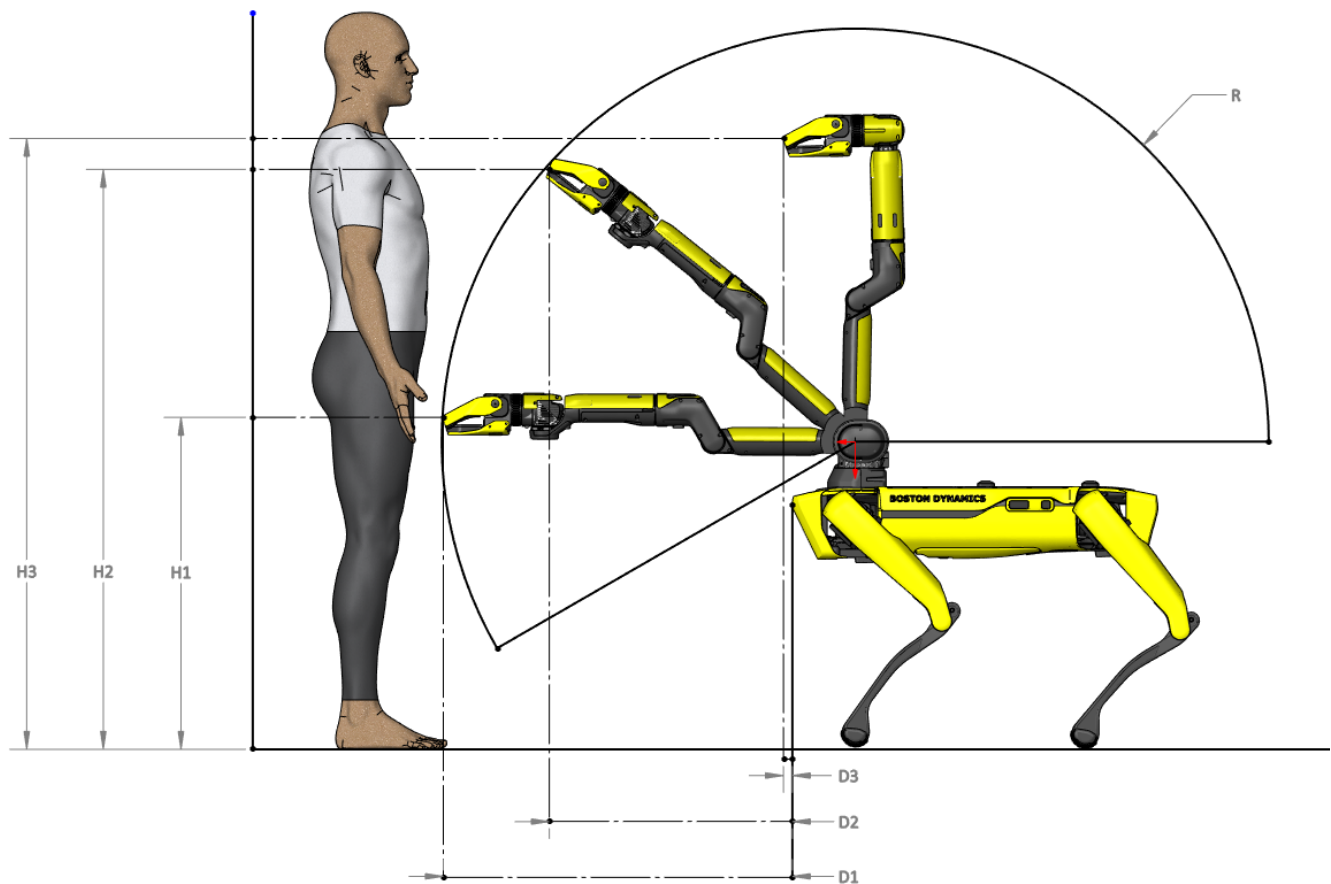
When performing the risk assessment, consider:

- The severity of hazards associated with pinch points in the legs and hips can be moderate or high, particularly for hands and fingers.
- Pinch points along legs or manipulator joints are most likely accessible in powered off conditions, at startup, or during maintenance and handling.
- Pinch points are likely to be determined by the size and geometry of manipulated parts.
- Pinch points between the manipulator elbow and potential payloads can occur, depending on the sizes of payloads.



2.4.3 Estimating Spot Arm workspace extension

The workspace of Spot Arm extends approximately 1 m beyond the footprint of the base of Spot.



Spot + Spot Arm workspace dimensions

The highest point of reach combines the height of the quadruped robot unit and the elevation angle of the manipulator.

The principal figures for estimating the extension of the Spot + Spot Arm workspace are reported in the following table.

Dimension	Description	Typical value/range	Notes
R	Maximum extension of Spot Arm	985 mm	At the tip of the gripper
H1	Height of Spot + Spot Arm in fully outstretched horizontal configuration (maximum horizontal extension of Spot Arm)	1100 mm +/- 160 mm	Variable depending on height and pitch/roll of Spot unit
D1	Extension of Spot Arm beyond the footprint of Spot, in fully outstretched horizontal configuration (maximum horizontal extension)	850 mm	Furthest point forward
H2	Height of Spot + Spot Arm in 45° configuration	1480 mm +/- 160 mm	Variable depending on height and pitch/roll of Spot unit
D2	Extension of Spot Arm beyond the footprint of Spot, in 45° configuration	580 mm	Combination of moderate extension and high elevation
H3	Height of Spot + Spot Arm in fully vertical configuration (minimum horizontal extension)	1560 mm +/- 160 mm	Variable depending on height and pitch/roll of the base
D3	Extension of Spot Arm beyond the footprint of Spot in fully vertical configuration (minimum horizontal extension)	20 mm	



NOTICE

When performing the risk assessment, consider:

- Pitching of Spot is generally limited to $\pm 30^\circ$ in typical configurations. Pitching can compensate for or increase the total elevation angle of Spot Arm relative to its mounting. This configuration affects the values of H and D dimensions, as shown in the table above.
- Elevation angles higher than 60° will maintain the workspace of Spot Arm very close to the footprint of Spot.
- Spot + Spot Arm is not likely to reach the upper human body for elevation angles that are commonly used for moving Spot Arm of the Spot footprint.

2.4.4 Estimating the risk of unexpected movements during manipulation

Operating and practicing with Spot + Spot Arm may produce unexpected movements in any direction.

The fastest movements are correlated with the swing of the manipulator around the vertical axis exiting from the shoulder joint with the manipulator fully extended. This configuration is rarely used for manipulation and locomotion.

NOTICE

When performing the risk assessment, consider:

- The main direction of motion of the base robot and the direction of motion expected for manipulation.
- The extended workspace of Spot + Spot Arm, which is the combined workspace of the manipulator and the displacement of Spot.
- Any object or equipment within the operating range of the Spot + Spot Arm. Make sure to operate in an environment with sufficient clearances and low probability of contact with equipment.
- The nature of the parts manipulated. Loss of grasp may generate instantaneous forces and

	sudden recovery movements, potentially leading to the loss of stability.
--	--

2.4.5 Estimating the risks associated with manipulation and balancing

The robot's upright and/or balancing posture can only be attained with active control. However, active control does not ensure that stable balancing is attained.

Ground conditions are not entirely predictable — Spot + Spot Arm could fall unexpectedly.

Touching objects or equipment with Spot Arm may affect the overall balance of Spot + Spot Arm. Spot + Spot Arm may react with large movements to compensate for external forces or the onset of imbalance.

	When performing the risk assessment, consider: • That the occurrence of loss of stability in the robot increases when Spot + Spot Arm is used to manipulate objects or equipment on elevated or inclined terrain. • The effect of the offset in the center of mass due to the configuration of the manipulator. • The combined effect of the movement of the arm, expected acceleration of the base, external loads and the weight of parts manipulated.
--	---

	Refer to <i>Spot Instructions for Use</i> on the Boston Dynamics Support Center for more information.
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2.4.6 Estimating the risks associated with navigation

The Spot + Spot Arm is intended to navigate in an environment with the manipulator in Stowed pose.



Refer to *Spot Information for Use* on the [Boston Dynamics Support Center](#) for details about safe navigation.

Moving Spot Arm during robot navigation may affect the balancing capability of Spot + Spot Arm, resulting in potentially large displacements around the expected trajectory, fast recovery of occasional unbalances, and rapid compensation movements.

Such movements are not necessarily an indication of failures or imminent loss of stability.

	When performing the risk assessment, consider: • The presence of stairs, inclines, and open edges • The presence of elevated grounds and the relative location of people with respect to Spot + Spot Arm • The presence of narrow passages that may be too small for the manipulator in any configuration different from the stowed position • The possibility to occasionally share the path with Spot + Spot Arm and travel direction of Spot + Spot Arm
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	Spot + Spot Arm could fall from stairs or elevated grounds, and cause personal injuries or material hazards. Do not stand below Spot + Spot Arm. When manually driving Spot + Spot Arm: • Avoid turning on stairs • Do not attempt to climb stairs by side-stepping • Navigate the robot to a complete landing and secured zone before turning
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2.4.7 Estimating the risks associated with collisions

Obstacle detection and avoidance applies only to the base robot. Objects along the trajectories of the manipulator part are not detected.

Each joint of the manipulator has a mechanical torque limitation. External loads due to impacts will make joints slip until an external constraint or hardstop activates to prevent further slips. See [Torque limitation](#).

	When performing the risk assessment, consider: • The possible configuration of the manipulator along a trajectory of potential contact. The impact force may depend on both the trajectory of Spot Arm and the walking path of Spot. • The programmed trajectory of the manipulator including large displacement or rotations around the shoulder joints, which determine large sweeping volumes. • The actual external torque loading each joint of Spot Arm in case of contact. • The maximum joint velocities of the manipulator. • The likelihood of manipulating while navigating, and the starting configuration of the manipulator if not stowed. • The velocity of the robot base, which can be up to 1600 mm/s.
---	---

2.5 PPE

While operating Spot + Spot Arm, no additional PPE is needed other than what is required in the User's industrial environment. However, wearing safety footwear is recommended.

2.6 EMC considerations

Spot and Spot Arm have been tested for EMC compliance according to standards related to heavy industrial environments. Spot should be able to withstand EMC disturbances typically found in Industrial environments.

See section [Compliance statement](#) for details.



2.7 Laser

Spot + Spot Arm has the following camera systems:

- Five stereo camera systems on the robot base that contain a Class I laser projection system.
- One camera system inside the gripper that contains a Class I laser projection system.



This product is classified as a Class 1 Laser Product under the EN/IEC 60825-1, Edition 2 (2007) and Edition 3 (2014).



Tampering with or adjusting any of the laser components is not allowed and may result in hazardous radiation exposure. Use of controls or adjustments or performing procedures other than those specified herein may also result in hazardous radiation exposure.

2.8 General warnings



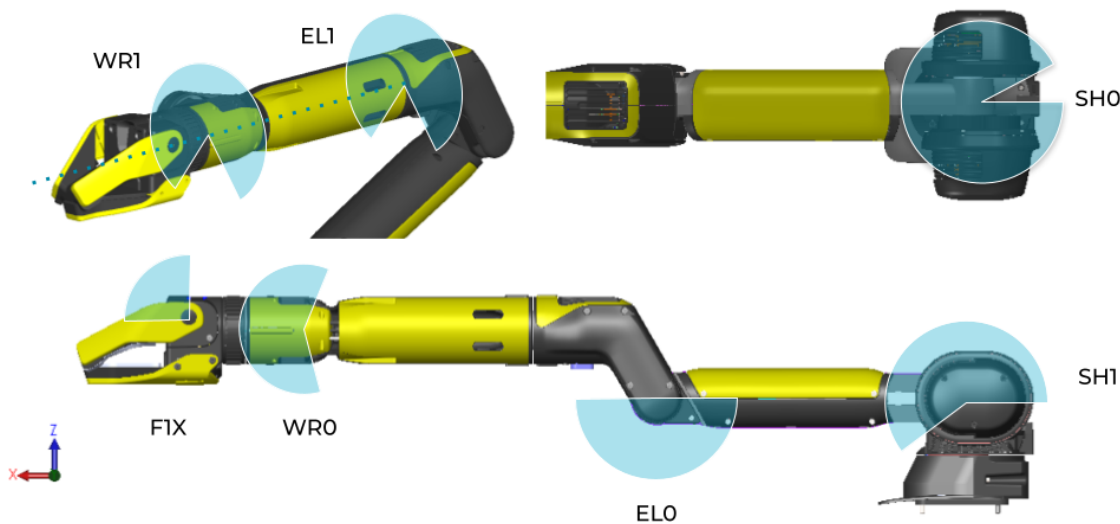
- Do not try to troubleshoot manipulation failures or malfunctions by touching the manipulator part.
- Do not attempt to access the gripper when motors are powered on.
- Do not attempt to manually intervene to help Spot + Spot Arm balance by touching the base or the manipulator.
- Make sure that there is sufficient clearance during training and setup to avoid contact with people and objects or fixtures due to unexpected movements.
- Do not stay on either side of Spot + Spot Arm during teaching or setting tasks.

3 Safety features

3.1 Torque limitation

Manipulator joint torque is mechanically limited as indicated in the following table:

Joint Axis	Max torque
SH0	90 Nm
SH1	90 Nm
EL0	20 Nm
EL1	20 Nm
WR0	20 Nm
WR1	20 Nm



Spot Arm joint axis references



Exceeding the maximum torque will make a joint slip until a hardstop is reached. Motors will quickly reduce power to prevent overloading the slipping joint.

The value of the maximum torque may be used to estimate the maximum force on potential impacts or static contacts.

NOTICE

Spot Arm has an additional software force limiter to reduce the external forces to the base (through the manipulator) that may cause loss of balance. This is not a safety function.

3.2 Stopping functions

Stopping Spot Arm is similar to stopping the Spot robot.



Refer to *Spot Instructions for Use* on the [Boston Dynamics Support Center](#) for more information about safe stopping Spot.

3.2.1 Manual stop using the controller

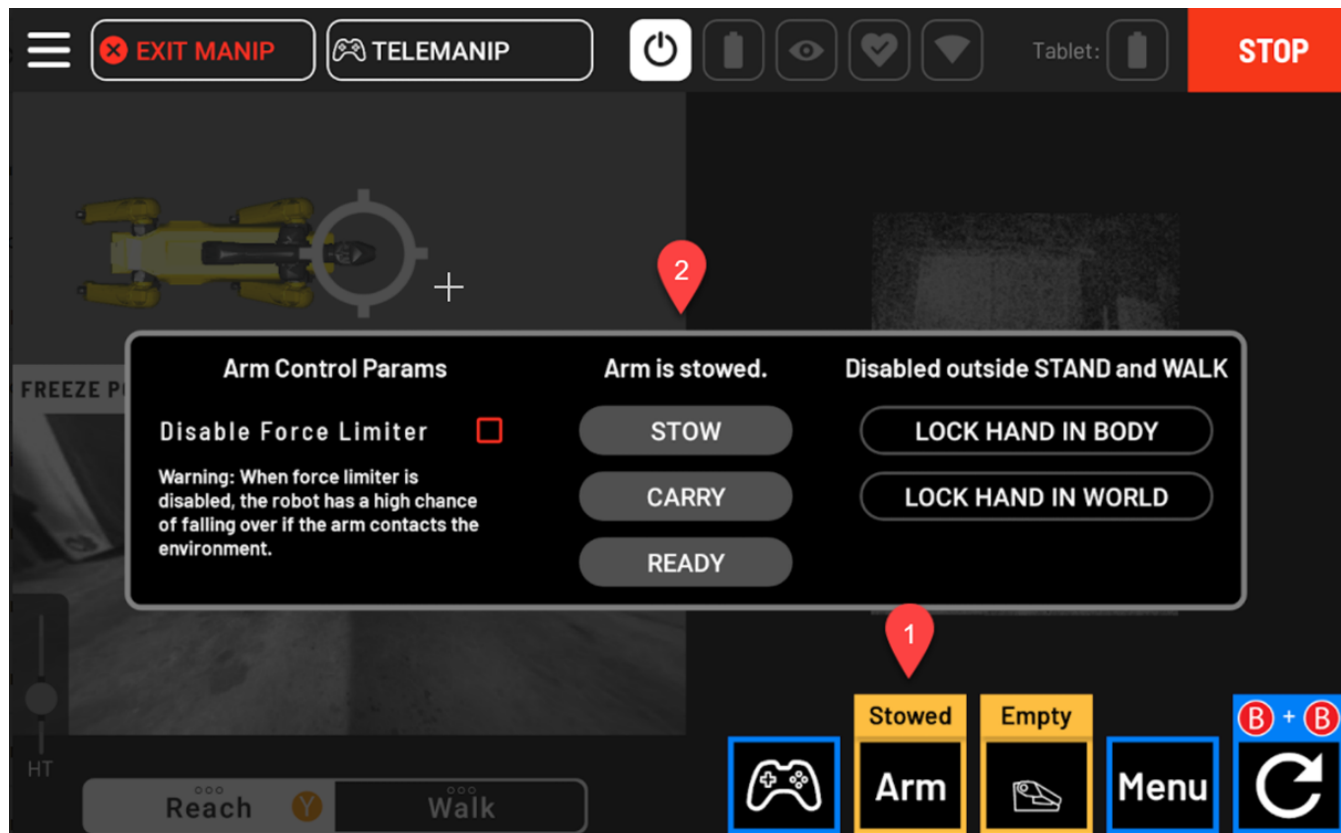
To suspend robot motion, press the B button on the controller to stop Spot + Spot Arm. The robot will pause and stand in place. If Spot is seated, it will remain seated.

To de-energize the robot motors:

1. Press the controller trigger buttons and the B button simultaneously.

OR

2. Select the red STOP button in the controller's top right corner. The robot will immediately sit down.



When STOP is commanded and confirmed, it takes precedence over all active commands on the controller. Motors cannot be turned on until STOP is canceled.

At any time during Spot Arm operations, Users can stop executing the current manipulator action and return the manipulator to the Stow position by pressing B twice (B + B) on the controller.

3.2.2 Protective stop

Spot + Spot Arm can be stopped by interfacing an external safeguard signal to the payload port. The stopping function will de-energize all motors (legs and manipulator).

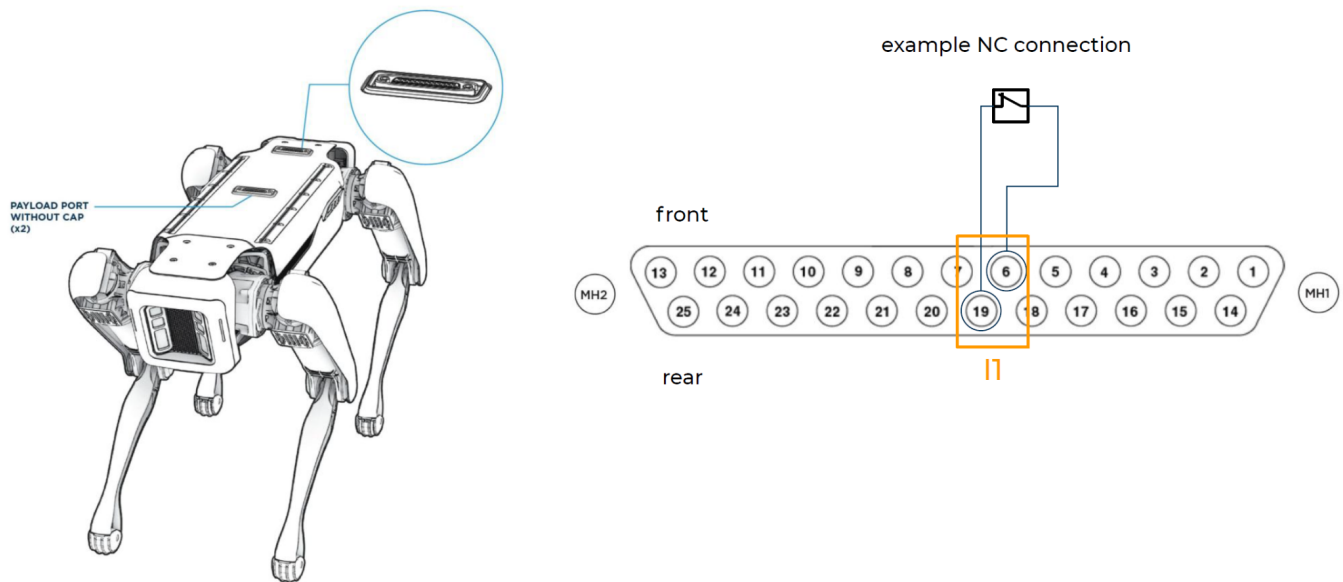


	When activated, the stopping function will cause the robot to lose its ability to stand and balance. The Spot Arm will NOT keep its joints locked. On flat ground, Spot + Spot Arm lowers its body and the manipulator collapses on top of Spot. On inclined surfaces or stairs, Spot + Spot Arm may tip-over. See Estimating the risks associated with manipulation and balancing.
---	---

Inputs for activating and triggering stopping functions are integrated with the robot's payload ports, as described in the following table.

Item	Description
Safe input location (see Figure)	I1 = pin 19
Interface	Use only one port, either front or back. See example in Figure below. Spot will not work without a cap or a properly configured payload attached to each port. When a cap is used, the stopping function is disabled.
Safe input default connection.	6-19 pair normally closed (NC).
Type of stop	Stop Category 0 (IEC 60204-1) when contact on I1 is opened.
Stopping safety function	Implemented in accordance with ISO 13849-1:2016, of Category 1 and performance level c. The PFHD is 1.1E-6/hr.
Reset	6-19 pair is closed.
Restart	Enable power following commands on the manual controller or issued from programs.
Environment	The safety-related part of the control system operates within the same environment limits established for the robot.

Refer to [Spot robot electrical interface](#) in the [Boston Dynamics Support Center](#) for details about payload ports.



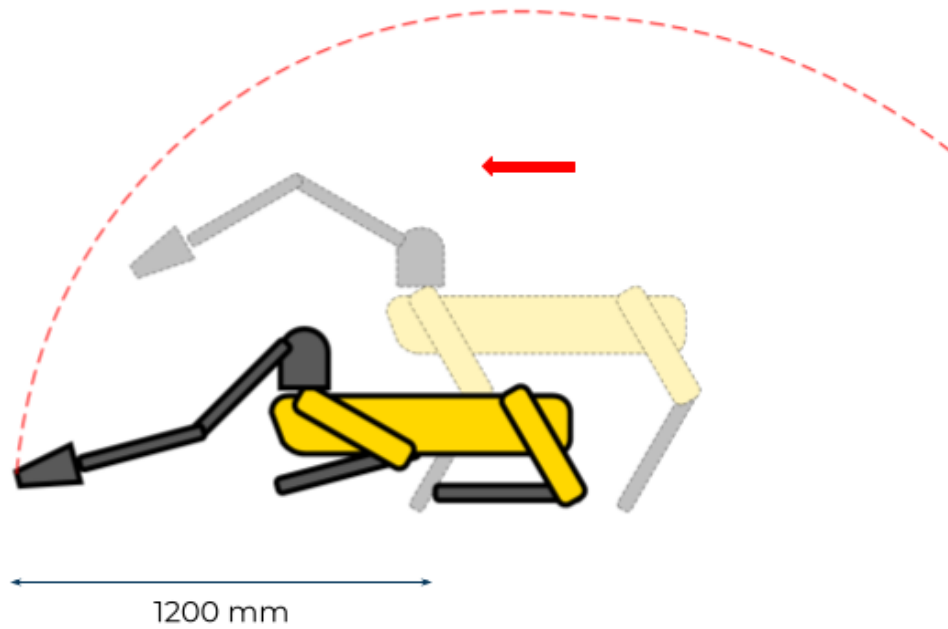
Payload port external signal interface for protective stop

3.2.3 Stopping time and distance

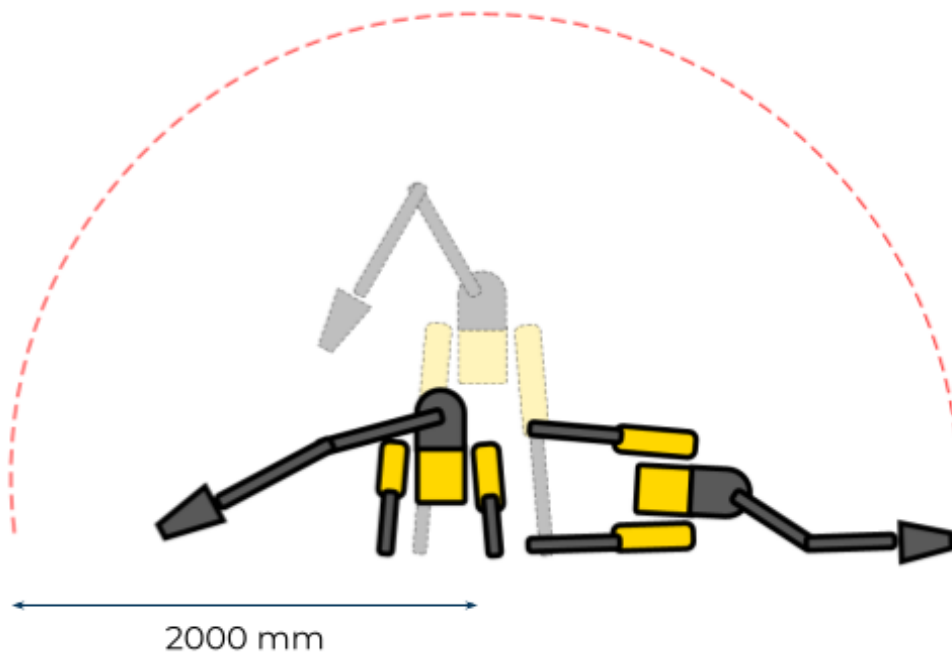
The maximum response time for the stopping function is 200 ms.

The de-energized safe state after a stop results from the Spot + Spot Arm gradually or suddenly going into the ground, with all motors powered off. The configuration of legs and manipulators is not predictable, as it depends on the dynamics of the stop.

Allow the following minimum distances around the robot for completing a stop.



Minimum stopping clearance forward and back



Minimum stopping clearance laterally

4 Transport, handling, and storage

4.1 Transport

Spot Arm is transported in two customized cases designed exclusively for the robot and its associated components.

NOTICE	Do not discard the cases after removing Spot + Spot Arm. The cases are designed for transporting the robot and its accessories, and for returning them to Boston Dynamics for service.
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Shipping case for the robot Length: 1060 mm Width: 552 mm Height: 378 mm Empty container: 19.3 kg Combined weight: 55.7 kg	Shipping case for battery pack/power supply Length: 559 mm Width: 432 mm Height: 267 mm Empty container: 9.2 kg Combined weight (one battery): 18.1 kg Combined weight (two batteries): 22.7 kg
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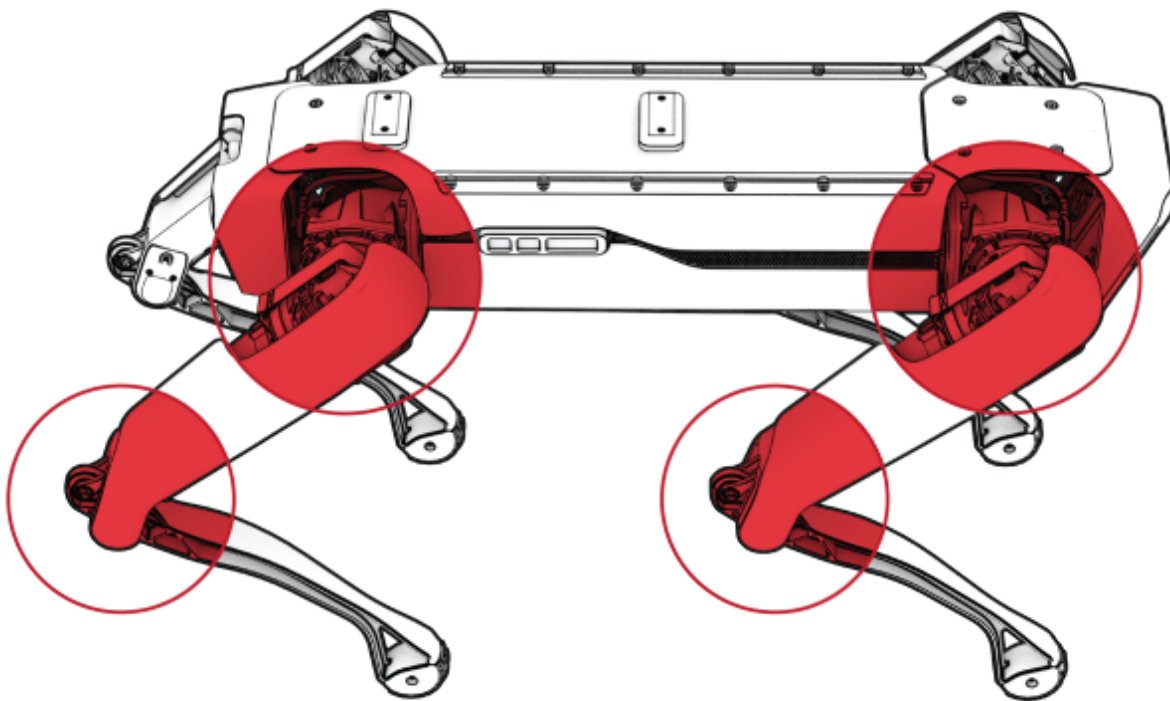
4.2 Safe handling

Refer to the Boston Dynamics Support article [Spot Arm boxing and unboxing](#) for information about moving the manipulator part of Spot Arm during boxing and unboxing procedures.



Spot Arm must be manually configured in Stow position before lifting Spot + Spot Arm.

Joints in the legs can pinch fingers and other body parts and entangle loose clothing, long hair, and jewelry.



Pinch point risk zones



When lifting Spot + Spot Arm, the manipulator may move out of its stowed position if the robot is tilted during the lifting process. Two people are required to lift Spot + Spot Arm.

Spot has a handle at each hip joint. Use these to lift, carry, and roll the robot.



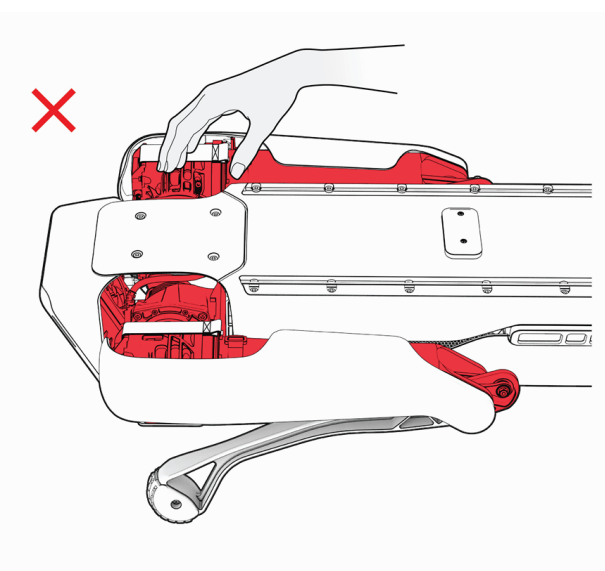
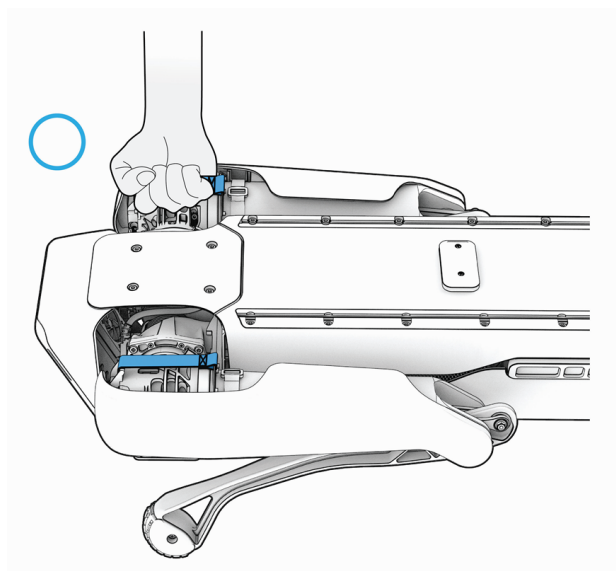
When using the handles, hands and fingers can be pinched.

Always keep hands away from knee joints. Use caution when lowering Spot Arm or closing the leg.

Always keep hands away from hip joints, except when grabbing handles. Always make a fist when gripping handles to keep fingers away from pinch points.

Always make a fist with each hand. Do not extend the fingers to avoid potential pinch hazard.

To avoid pinch points, never carry Spot + Spot Arm with fingers extended.

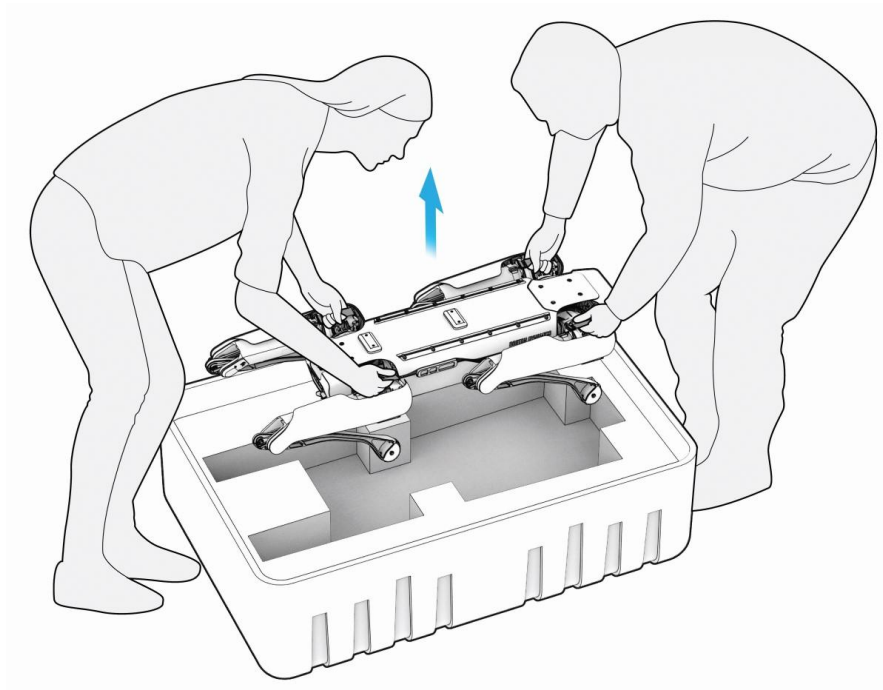


Use grasp handles to avoid pinch points



4.2.1 Proper two-person lift

Spot + Spot Arm combined weighs 40 kg and can be awkward to carry, particularly with attached payloads. Grab the handles and maintain safe lifting precautions. To lift Spot + Spot Arm, two people will need to assist, with one at the front and one at the rear.



Two-person lift using grasp handles

4.2.2 PPE

While handling Spot + Spot Arm, no additional PPE is needed other than what is required for operators in their industrial environment. However, it is recommended to wear safety footwear.

4.3 Storage

Store Spot + Spot Arm in a dry location that has adequate temperature controls. Spot + Spot Arm can be safely stored in temperatures between -30°C and 40°C, with relative humidity between 30% and 70%.

NOTICE

When not in use, remove the battery pack from Spot. Boston Dynamics recommends storing Spot + Spot Arm in the transportation case provided with the robot.

4.3.1 Battery storage

- Store batteries at -30°C to 25°C. Charge at 0°C to 40°C. Operate at -20°C to 45°C.
- Users should develop a battery storage and charging safety policy consistent with industry standards and local regulations.
- Remove the battery from the robot during transportation or storage.



Details and illustrated procedures are collected in a series of training videos on the [Boston Dynamics Support Center](#).



5 Installation and commissioning

5.1 Preparation

Before setting up Spot + Spot Arm, ensure that enough clearance is provided around the robot to ensure safe operation.



Check that a separation distance of at least 3 m from Spot + Spot Arm is established before commissioning for the first time.

5.2 Battery setup and charging

Spot + Spot Arm is powered by a removable lithium-ion battery pack. The battery can be charged standalone using the robot's battery charger or while installed in the robot.

The robot's battery must be removed from the robot when not in use.

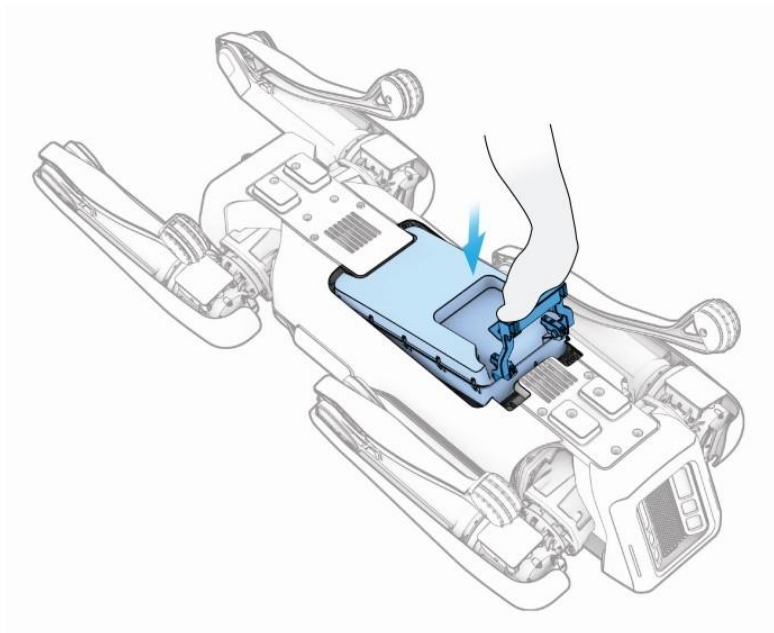


Do not short-circuit, burn, disassemble, submerge, puncture, crush, drop, or damage the battery.
If a battery fire starts, do not try to put it out. Evacuate to a safe area and contact your local emergency services. Battery fires create toxic fumes and cannot be put out with conventional fire extinguishers or water.

Charge the battery only using the charger provided by Boston Dynamics.

To install the battery:

1. Position the robot to expose the battery compartment.
2. With the battery compartment exposed, slide the battery into the battery compartment.
3. Latch the handle.
4. Ensure the battery is properly latched.



Inserting the battery

 DANGER	To reduce the risk of electric shock and fire: • Use a properly grounded outlet. Do not use ground adapters or a replacement plug. • Do not touch uninsulated parts of the output connector or battery terminals. • Do not open or disassemble the charger. • Do not use if the AC cord is damaged.
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Operate the charger in the following conditions:

- Charge at: 0°C to 40°C
- Input: 100-240 VAC, frequency 50/60Hz, Current 5.5A
- Max Charge time: 2.5 hours

	Details and illustrated procedures are collected in a series of training videos on the Boston Dynamics Support Center.
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6 Use of the machine

The Spot + Spot Arm can be operated in various manipulation modes.



Details and illustrated procedures are collected in a series of training videos on the [Boston Dynamics Support Center](#).

6.1 Preprogrammed poses: Stowed and Ready

Spot Arm has two predefined positions.



Stow pose



Ready pose

When a preset pose is requested by commands, the Spot Arm will quickly move to the final configuration.

Pressing B twice (B +B) on the controller will return the Spot Arm to the Stow pose, immediately halting the execution of the current action.

When performing the risk assessment, consider (among other factors):

- The effect of opening or closing the gripper
- The effect of parts, if manipulated or carried
- The trajectory of the manipulator
- Potential movements of the base robot for maintaining balance

Maintain a minimum clearance of 3 m around the robot.

6.2 Typical tasks

The following is a list of built-in programmed tasks or motion features. Refer to the Boston Dynamics Support Center for more information about using Spot Arm.

Arm task or position	Description
Reach	The Operator uses the controller joysticks to drive the end-effector.
Drive	The Operator navigates the robot with the tablet showing the selected camera view.
Choose camera input	The Operator chooses which camera to use: Robot cameras, gripper camera, or other attached camera or sensor payloads.
Follow	The robot body walks to follow the end-effector.
Locked hand in body	The manipulator joints are frozen and the manipulator remains fixed relative to the robot's base.
Locked hand in world	The end-effector (gripper) holds its position in the world as the body moves around it. ⚠ WARNING: Potential unexpected configurations or trajectories may occur.
Carry	The end-effector moves to a hard-coded position (similar to Ready) for carrying an object. ⚠ WARNING: While moving with the manipulator deployed forward, collisions with people and objects or fixtures can occur.



	Small rotations of the base will determine large rotations of the manipulator.
Twist, turn, and pull	Grasp and manipulate various controls, such as valve levers, wheels, and switches. ⚠ WARNING: Sudden loss of grasp may occur and result in unexpected movements of the manipulator and the base. ⚠ WARNING: Never perform manual intervention while the robot is being used and do not approach the robot. Never handle the gripper if it gets stuck during operation. Always power down the robot before handling any part of it.
Drag	The robot drags an object it cannot lift. ⚠ WARNING: In the event of loss of grasp during dragging, the Spot Arm may experience large compensation movements.
Door opening	Two door behaviors are available: Open Door and Open Grasped Door. ⚠ WARNING: Always stand clear of both sides of the door.

7 Maintenance

There are no user-serviceable components of the Spot + Spot Arm. Contact Boston Dynamics Support for help with maintenance.

7.1 Visual inspection of Spot Arm

Periodic inspection of the Spot Arm may identify damage that could generate hazardous conditions.

To conduct a visual inspection of Spot Arm:

1. Move the Spot Arm to the stowed position and check that Spot Arm configuration is completely folded.
2. Check that revolving joints do not present signs of misalignment, wear or damage.
3. Check that manipulator and gripper outer coverings do not present visible evidence of damage.

Do not attempt to repair the Spot or Spot Arm. Spot Arm is not meant to be repaired by Users.

Contact Boston Dynamics Support for help servicing Spot or Spot Arm by emailing support@bostondynamics.com. Please include the following information when contacting Support:

- Robot serial number
- Description of the issue

Any unauthorized repair or servicing of Spot will invalidate any representation or warranty made by Boston Dynamics with respect to machine safety.

Refer to *Spot Information for Use* on the [Boston Dynamics Support Center](#) for details about the Spot robot and Spot Arm.



8 Compliance statement

8.1 EU Declaration of Incorporation

Products in the scope of the Declaration of Incorporation:

- Spot + Spot Arm (a robot composed of a mobile unit equipped with 4 articulated limbs for locomotion and a multi-axis manipulator assembled on top of the mobile unit, controlled by a shared controller). The function is determined by the completed machine (application and intended use).

Boston Dynamics declares, under its sole responsibility, that Spot + Spot Arm, for what is supplied, shall only be put into service upon being integrated into a final complete machine (robot application) which must conform with the provisions of the Machinery Directive 2006/42/EC and amendments (MD) and other applicable Directives, and with the regulations transposing EU Directives into national law.

Compliance with all essential requirements of Machinery Directive 2006/42/EC relies on the correct incorporation conditions into a specific robot application and a final risk assessment.

The following Essential Safety and Health Requirements (EHSR) and supplementary requirements of Annex I of the Machinery Directive 2006/42/EC and amendments (MD), have been applied and fulfilled: 1.1.2, 1.1.3, 1.1.5, 1.2.1, 1.2.2, 1.2.3, 1.3.2, 1.3.4, 1.3.7, 1.5.1, 1.5.2, 1.5.6, 1.5.10, 1.5.11, 1.5.12, 1.7.1, 1.7.2, 1.7.3, 1.7.4, 3.5.1

It is declared that Spot + Spot Arm fulfils all the relevant provisions of the following directives:

- EMC Directive 2014/30/EU and amendments (EMCD),
- Radio Equipment Directive 2014/53/EU and amendments (RED)

Harmonized Standards Used, as referred to in:

- EMCD: EN 61000-6-4:2007/A1:2011, EN 61000-6-2:2005/AC:2005
- RED: ETSI EN 301 489-1 V2.2.1 (2019), ETSI EN 301 489-17 V3.2.2 (2019), EN 55032:2012

Other Technical Standards Used:

- EN ISO 12100:2010, EN 60204-1:2018,
- EN IEC 62133-2:2017, EN IEC 60825-1:2014/A11:2020

Boston Dynamics shall transmit relevant information about the partly completed machinery in response to a reasoned request by the national authorities.